

The Role of Urban Greenspace in Shaping Labor Market Outcomes

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Job Market Paper

November 10, 2025

Abstract

This paper examines how urban greenspace influences labor market outcomes, focusing on whether workers are willing to accept lower wages in exchange for greener environments. Using a panel dataset of 950 urban U.S. counties from 2011 to 2019, we apply the theory of compensating differentials within a spatial equilibrium framework. Greenspace is measured using the Normalized Difference Vegetation Index (NDVI), a remote sensing indicator derived from satellite imagery. To address endogeneity concerns, we employ an instrumental variables strategy that leverages historical land cover data from the 1970s-80s. Our findings show a significant negative relationship between NDVI and wages, suggesting that workers accept lower pay in greener counties due to the non-monetary benefits of greenspace. At the same time, we find that greener cities have higher employment and population levels, indicating broader economic appeal. Sector-level analysis further reveals that the strength of this wage effect varies by industrial composition, with more diversified economies exhibiting lower wage reductions. These results highlight the economic value of greenspace beyond environmental or aesthetic benefits. As an amenity that can be shaped through public policy, greenspace offers a tool for enhancing urban livability and attracting labor, making it a valuable component of sustainable urban development strategies.

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1 Introduction

Urban greenspace is often viewed as a desirable feature of sustainable cities, credited with improving public health (WHO, 2016), mitigating heat (Gunawardena et al., 2017), managing stormwater (Berland et al., 2017), and enhancing overall livability (Reyes-Riveros et al., 2021). However, in cities where land is scarce and development pressures are high, policy-makers and planners frequently face trade-offs between allocating land for greenspace versus using it for higher-yielding commercial or residential purposes. In such contexts, greenspace is typically treated as a secondary amenity, valued in principle but de-prioritized in practice. This study offers a concrete economic rationale for investment in urban greenspaces. By showing that workers are willing to accept lower wages in exchange for access to greenspace, it quantifies a compensating differential that can help justify urban greening efforts not merely on environmental or aesthetic grounds, but as a strategy for enhancing a city’s economic competitiveness and long-term appeal.

This study asks a simple but underexplored question: *do urban workers accept lower wages in exchange for living in greener areas?* In other words, is greenspace a local amenity that improves quality of life, leading people to accept lower incomes in return? According to the theory of compensating differentials (Roback, 1982; Rosen, 1979), equilibrium wages reflect not just productivity but also local amenity values. In equilibrium, utility is equalized across cities and if an amenity is utility enhancing, workers are willing to accept a lower wage as long as their overall utility remains constant. While such non-market valuation frameworks have been applied to value amenities like climate (Albouy et al., 2016), air quality (Chay and Greenstone, 2003; Currie et al., 2015), and crime (Gyourko and Tracy, 1991; Dentler and Rossi, 2024), the wage valuation of urban greenspace remains largely unexplored in the economic literature.

To examine the utility value of greenspace, we use a panel dataset covering 950 urban counties in the United States (U.S.) from 2011 to 2019. Our measure of greenspace is the Normalized Difference Vegetation Index (NDVI), a remote-sensing index derived from

satellite imagery that captures the extent and density of green vegetation, widely used in other settings (Huang et al., 2021; Zheng et al., 2021; de la Iglesia Martinez and Labib, 2023; Davis et al., 2023). NDVI offers high spatial and temporal resolution, allowing for consistent, comparable tracking of greenspace across counties over time. We combine this with county-level wage data from the U.S. Bureau of Labor Statistics (BLS) and a rich set of controls, including measures of rents, employment structure, and socio-economic characteristics, to estimate the relationship between greenspace and wages. To address potential endogeneity concerns, we employ an Instrumental Variable (IV) strategy using historical land cover data from the 1970s-80s, leveraging the persistent nature of land use patterns to isolate exogenous variation in contemporary greenspace.

Our results indicate a robust and statistically significant negative relationship between greenspace and urban wages, consistent with the hypothesis that workers value greener environments and are willing to trade off earnings to access them. In our preferred model specification, a 1 standard deviation increase in NDVI is associated with a 3.6% decrease in average annual wages, holding other factors constant. We also find that greener cities have higher population growth and more employment, factors important to city growth, compared to their less green counterparts.

By documenting the wage-amenity tradeoff associated with urban greenspace, this paper makes several contributions to the literature. First, it extends the compensating differentials framework to a widely recognized but understudied amenity. While most prior studies on greenspace rely on perceived measures derived from qualitative surveys (Lafortezza et al., 2009; Kabisch, 2015), we take a revealed preference approach, using wage variation as an implicit valuation mechanism. Second, while much of the environmental economics literature focuses on housing prices as the primary outcome (Cho et al., 2008; Geoghegan, 2002; Zambrano-Monserrate et al., 2021; Zhang et al., 2020), our wage-based estimates provide an independent and theoretically consistent measure of how greenspace enters the utility function and influences labor market dynamics.

In addition, to the best of our knowledge, this is the first paper to take a causal approach to estimating the value of greenspace using a compensating differentials framework. By leveraging historical land cover as an instrument for contemporary greenspace, we address endogeneity concerns such as reverse causality and omitted variable bias, issues that often limit interpretation in this area. Finally, unlike many natural amenities such as coastline or elevation, greenspace is malleable through policy. Urban tree planting, park creation, and land use regulations can significantly alter greenspace availability within relatively short time frames, as opposed to the availability of a coastline or a river. As such, our findings offer actionable insights for planners and policymakers seeking to improve urban livability and attract or retain labor through environmental interventions.

The remainder of this paper proceeds as follows. The *Background* section discusses the previous work done in this context and explores the role of natural amenities like greenspace. The *Data* section gives an overview of the data used in the analysis. The *Methodology* section describes the theoretical background, the empirical formulation of the hypothesis and details the identification strategy. The *Results* section discusses the empirical findings, and finally the paper ends with the *Conclusion* section.

2 Previous Literature

The spatial-equilibrium and compensating-differentials framework developed by Roback (1982) and Rosen (1979) provides the standard lens for valuing local attributes. In this framework, wages and housing costs adjust so that, in equilibrium, utility is equalized across space. This approach underpins a large literature that infers amenity values from observed wage-rent patterns. Many studies have focused on housing and land markets and other related fields (Anenberg and Kung, 2020; Harris, 2024; Glaeser et al., 2014; Bischoff, 2012; Kahn and Tracy, 2024). A central implication of the model is that both labor and housing outcomes are informative about the relative desirability of places, with the partition across wages

and rents depending on the extent to which amenities shift residential utility, production conditions, or both (Roback, 1982, 1988).

A complementary literature, tracing back to the 1970s, (Polinsky and Shavell, 1976; Yinger, 1976) conceptualizes cities as bundles of consumption opportunities. In this view, households choose locations partly to access amenity sets that are difficult to replicate privately. Glaeser et al. (2001) formalize the idea of the “consumer city,” and Glaeser and Gottlieb (2006) emphasize how advances in transportation and communication loosened the primacy of proximity-to-jobs, increasing the salience of consumption amenities such as cultural institutions, varied dining, entertainment, school quality, and neighborhood safety. This body of work uses inter-city comparisons to show that amenity differences help explain where people choose to live and, through equilibrium adjustments, why wages and rents differ across places. In short, amenities are not peripheral—they are central to urban sorting and to the equilibrium prices that studies use to infer quality-of-life (QOL) differences (Glaeser et al., 2001; Glaeser and Gottlieb, 2006).

Environmental attributes are a prominent component of the amenity bundle examined in QOL studies. Climatic factors, water access, and resource endowments have been analyzed within compensating-differentials frameworks to explain systematic regional differences in outcomes (Albouy et al., 2016). Recent work highlights how natural amenities and resource settings can shape development trajectories and location choices, reinforcing the role of environment in spatial equilibrium (Clay et al., 2023). At the urban scale, the presence of natural amenities is linked to growth by improving a city’s ability to attract households and firms (Rickman and Wang, 2017; Stephens and Partridge, 2015; Hausman et al., 2025; Weinstein et al., 2023). This environmental strand thus connects directly back to the QOL literature, when environmental conditions raise residential utility or affect production, equilibrium wages and rents adjust in ways that reveal the value of those conditions (Roback, 1982; Rosen, 1979).

Within environmental amenities, urban greenspaces like parks, tree-lined streets, gar-

dens, and other associated features, have been studied extensively for their contributions to physical and mental health, air-quality exposure, and social well-being (Carrus et al., 2015; Wolf et al., 2020; Donovan and Gatziolis, 2019; Weichenthal et al., 2016; Gu et al., 2022). Greenspaces can also function as “third places,” facilitating interaction and community ties (Kabisch, 2015), and they provide non-market benefits that are salient to household location choices (Lafortezza et al., 2009). Empirically, much of this literature documents capitalization of greenspace into housing prices and land values (Siriwardena et al., 2025; Pandit et al., 2014), and evaluates broader cost–benefit implications of urban trees and related investments (Soares et al., 2011). Under the spatial-equilibrium lens, greenspace is treated as a consumption amenity, shifting residential utility rather than firm productivity. Unlike production-linked attributes such as access to waterways, greenspace is also primarily shaped by urban policy. Under that interpretation, compensating differentials may manifest in both wages and rents consistent with Roback (1982, 1988) and more recent literature (Chen and Wang, 2013).

Taken together, the QOL, amenities, and environmental-economics literatures establish that local attributes, especially environmental ones, have measurable value revealed in equilibrium prices and quantities (Roback, 1982; Rosen, 1979; Albouy et al., 2016; Weinstein et al., 2023). However, within the greenspace literature specifically, the emphasis has largely been on health and social outcomes and on housing-market capitalization (Carrus et al., 2015; Wolf et al., 2020; Donovan and Gatziolis, 2019; Kabisch, 2015; Siriwardena et al., 2025; Pandit et al., 2014; Soares et al., 2011). By contrast, direct evidence on the contribution of urban greenspace to wage differentials remains comparatively limited. Clarifying that wage channel within the established spatial-equilibrium framework is the natural next step implied by this literature.

3 Data

Our analysis uses data on 950 urban counties in the contiguous 48 U.S. states (excluding Alaska and Hawaii) the years 2011 - 2019. For classification of urban and rural, we rely on the identification provided by the Office of Management and Budget (OMB, 2023).

The academic literature has defined greenspaces in various ways. A systematic review of the broad academic literature across multiple disciplines was conducted by Taylor and Hochuli (2017) and finds two major definitions of greenspace. For purposes of this study, we focus specifically on urban vegetation, encompassing features such as trees, parks, gardens and yards typically referring to vegetated forms of open space within urban environments (Chong et al., 2013; Bastian et al., 2012). This is also consistent with the definition from the U.S. Environmental Protection Agency (EPA) which defines greenspace as urban land that is partly or completely covered with grass, trees, shrubs, or other vegetation (EPA, 2023).

We measure greenspaces using the MODIS NDVI product (MCD13Q1 v061), which provides 16-day composites at 250-meter resolution from the Terra and Aqua satellites. For each year 2011–2019, we mosaic the June–July scenes over the contiguous U.S., apply the product’s quality flags to mask clouds/snow, and reproject to NAD83 / CONUS Albers (EPSG:5070). We then aggregate NDVI to our analysis unit of counties to construct a consistent, area-appropriate panel of summer vegetation greenness. *Figure 1* shows a visual representation of what NDVI identifies as greenspace, highlighted in red, from the satellite image of Point State Park in Pittsburgh, Pennsylvania.

Although there are many vegetation indexes, the NDVI is preferred by many disciplines as it enables comparison across spatial and temporal spectrums (Huang et al., 2021) and thus has been widely used in analyzing greenspaces in other disciplines (Donovan and Gatzliolis, 2019; Huang et al., 2021; Lei et al., 2021; Davis et al., 2023; Zheng et al., 2021). The NDVI uses the Near Infra-Red band and the red band spectrums of satellite images to calculate an index value with a range of -1 to +1, with the higher the value, the higher the greenspace, values below 0 are associated with non-greenspaces (*equation 1* shows the

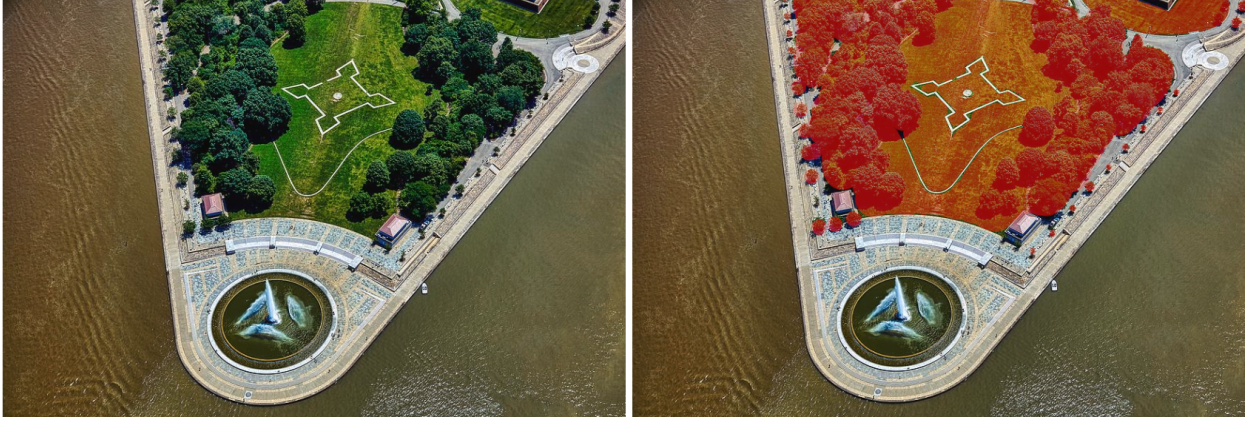


Figure 1: NDVI Vision

calculation formula).

$$\text{NDVI} = \frac{\text{NIR} - R}{\text{NIR} + R} \quad (1)$$

The satellite images were read into a GIS software, and were superimposed on a county shapefile. Using a raster extractor, the county level NDVI values were extracted as a mean of the all the NDVI indexes value grids within a county shape polygon. The previous literature also shows that NDVI values can fluctuate seasonally, however, it is the highest during the summer (Jia et al., 2004; Kulenbekov et al., 2021). At the same time, satellite images can be often compromised by cloud covers, making a one-off assessment faulty. As a result, to create our annual measures of greenspace, we used the bi-weekly satellite images for the months of June and July for each year while also correcting it for cloud or snow covers. In addition, our analysis exclusively considered values exceeding 0 when extracting data from the raster files. Our sample analysis reveals that the mean daily average NDVI value is 0.343 per county per annum, while the mean daily maximum value is 0.848 per county per annum, as depicted in *table 1*.

Figure 2 illustrates the mean NDVI values at the county level for the counties in the 48 contiguous U.S. states for the years 2000 and 2022, with more green areas representing a higher NDVI value, thus, a higher greenspace. It is also evident from the maps that

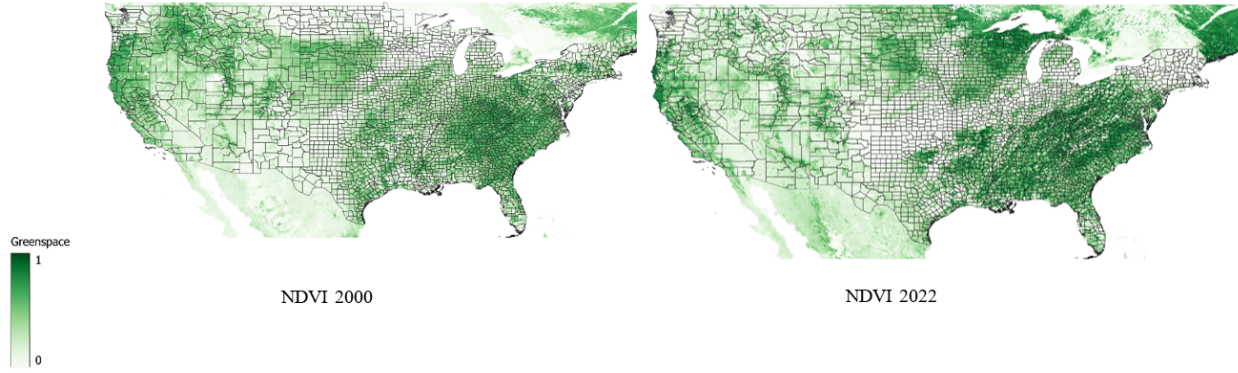


Figure 2: NDVI distribution across counties 2000 and 2022

greenspace changes over time and thus greenspace is not the same as the climate.

As described below, the primary dependent variable in our analysis is county-level wages, obtained from BLS. To examine broader labor market dynamics, we also analyze two additional outcomes: total population, sourced from the U.S. Census Bureau, to evaluate whether greenspace attracts more residents; and total employment levels, also from BLS, to assess whether greener areas exhibit higher employment. *Table 1* presents descriptive statistics for all variables included in the model. On average, counties report annual wages of approximately \$42,000.

Table 1: Descriptive statistics of key variables.

Variable	N	Mean	SD	Min	Max
Average Annual Wage	10,750	41,721	10,339	20,968	138,273
Annual average of daily mean NDVI	10,750	0.343	0.142	0.00481	0.848
Annual average precipitation	10,750	0.111	0.0387	0.00250	0.282
Annual High Temperature days	10,750	30.66	16.97	0	105
% of people employed	10,750	57.41	6.92	21.40	77.60
% of population with Highschool Degree	10,750	62.16	8.038	19	81.20
% of population with Bachelors Degree	10,750	15.84	5.955	2.600	38
% of population with Graduate Degree	10,750	8.986	4.755	0.400	40.30

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Variable	N	Mean	SD	Min	Max
% of population less than HS	10,750	13.01	5.812	1.300	49.50
Total Population	10,750	238,739	516,492	709	1.011e+07
% of people White	10,750	82.14	14.41	12.60	99.80
% of people Black	10,750	9.949	12.59	0	78.50
% of people American Indian/Alaskan	10,750	0.798	3.191	0	83.90
% of people Asian	10,750	2.161	3.263	0	36.50
% of people Hawaii/Pacific	10,750	0.0752	0.179	0	2.300
% of people other race	10,750	2.495	3.624	0	38.40
% of people multiracial	10,750	2.374	1.399	0	18.80
% of people working in-county	10,750	59.64	19.94	11	98.50
Miles to nearest MSA	10,750	15.79	12.70	0	62.15
Median Rent	10,750	916.46	266.15	497	3269
House Price Index (HPI-86 base prices)	10,539	363.41	240.56	79.36	2575
County land area (sq km)	10,750	2.194e+09	3.438e+09	5.868e+07	5.195e+10
NAICS_11	10,140	2.721956	4.805812	0	68.58669
NAICS_21	10,140	0.6705086	2.413327	0	57.22946
NAICS_22	10,140	0.5519818	1.224466	0	24.88364
NAICS_23	10,140	6.324081	2.794917	0.2606158	44.29266
NAICS_31	10,140	1.879054	3.225592	0	40.32463
NAICS_32	10,140	2.979599	3.085709	0	30.57712
NAICS_33	10,140	5.257561	5.236238	0	64.09611
NAICS_42	10,140	3.054581	1.648293	0	20.899
NAICS_44	10,140	5.094034	1.472625	0	13.71592
NAICS_45	10,140	5.970427	1.998292	0.0308603	37.42886
NAICS_48	10,140	2.527547	1.957322	0.046642	33.96606
NAICS_49	10,140	0.9493777	2.055554	0	55.95381
NAICS_51	10,140	1.094488	0.8930219	0	9.724973
NAICS_52	10,140	2.993985	2.229393	0.0004737	41.56274
NAICS_53	10,140	1.216823	0.6523989	0	8.1391

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Variable	N	Mean	SD	Min	Max
NAICS_54	10,140	3.952043	3.461146	0.112946	81.84882
NAICS_55	10,140	0.8352158	1.292226	0	29.70073
NAICS_56	10,140	4.722905	2.19046	0.1653116	21.49744
NAICS_61	10,140	1.736387	2.136355	0	35.575
NAICS_62	10,140	11.50715	4.487855	0.1153433	50.20877
NAICS_71	10,140	1.564459	2.089524	0	47.60313
NAICS_72	10,140	8.063525	3.585911	0	70.47432
NAICS_81	10,140	5.441253	1.48685	0.3166605	29.79453
NAICS_90	10,140	18.837	8.280837	0.2606457	89.59314
NAICS_99	10,140	0.0540546	0.1146866	0	1.167113
Hist_NAICS_7010	10,140	67182.26	196605.2	296	3391725
Hist_NAICS_7020	10,140	59286.09	180298.8	205	3022557
Hist_NAICS_7040	10,140	7896.164	17658.47	70	369168
Hist_NAICS_7050	10,140	975.4112	836.0193	0	8467
Hist_NAICS_7060	10,140	6920.752	17618.17	30	367024
Hist_NAICS_7070	10,140	1612.758	1923.212	0	25156
Hist_NAICS_7080	10,140	65569.5	196380.9	221	3383801
Hist_NAICS_7090	10,140	53962.73	169040.4	133	2940851
Hist_NAICS_70100	9,830	353.3032	889.9014	1	17391
Hist_NAICS_70200	9,420	373.9947	1488.075	0	29495
Hist_NAICS_70300	10,070	3296.813	8025.062	1	128165
Hist_NAICS_70400	10,040	14992.67	46916.36	1	824507
Hist_NAICS_70500	9,740	3856.352	12901.65	5	189524
Hist_NAICS_70610	9,830	3495.74	14149.56	0	236258
Hist_NAICS_70620	10,060	10280.05	28400.24	5	521782
Hist_NAICS_70700	10,020	4932.3	19958.21	0	435497
Hist_NAICS_70800	9,870	13248.09	43975.67	3	765538
Hist_NAICS_70900	10,140	11606.77	29674.04	69	466706
Hist_NAICS_70910	10,140	2092.667	6181.246	2	72673

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Variable	N	Mean	SD	Min	Max
Hist_NAICS_70920	10,140	2344.918	7677.724	2	156803
Hist_NAICS_70930	10,140	7169.186	20514.44	62	394880

We incorporate a comprehensive set of county-level characteristics as control variables to account for potential confounding factors. We incorporate indicators such as educational attainment levels, percentage of people working in the county, total county population and racial composition, derived from the American Community Survey (ACS) 5-year estimates. To account for the economic structure of each county, we include variables representing the industrial composition based on shares of employment in two-digit North American Industry Classification System (NAICS). We also control for historical industrial composition from the 1970s-80s in our IV specifications. The control variables also include the county’s land area, proximity to the center of the Metropolitan Statistical Area (MSA), and annual high temperature days and annual precipitation data obtained from the NOAA.

Our analysis focuses exclusively on urban counties, effectively comparing conditions across urbanized areas. A key challenge in this context is that many U.S. urban counties encompass significant rural land, such as forests, farmland, or undeveloped areas, that may be detected by NDVI as greenspace. However, these areas are often not directly accessible or enjoyed by the urban population, potentially biasing our measure of greenspace. To address this concern, we include present-day land cover as a control variable in our model. This allows us to account for the influence of non-urban vegetation and isolate the effect of greenspace that is more likely to be experienced within the urban environment.

Finally, the Roback model considers both wages and rents. As such, we control for the median rent using the Fair Market Rents (FMR) collected from the U.S. Department of Housing and Urban Development. The FMR provides median rents across different size of housing, which we averaged for each county per year.

4 Methodology

4.1 Theoretical Foundations

This paper builds on the spatial equilibrium framework introduced by Roback (1982) to examine how urban greenspace influences labor market outcomes. In this framework, workers are assumed to be fully mobile and choose among locations by comparing bundles of wages, housing, and local amenities. Because utility must be equalized across space in equilibrium, differences in local quality of life are offset by adjustments in

either wages or rents.

Urban greenspace enters this setup as a local amenity that improves individual well-being but does not directly affect productivity. The implication is straightforward: workers may be willing to accept lower wages in greener locations, since the utility they derive from the amenity compensates for the reduction in income. In other words, locations with greater greenspace can sustain lower wages while still attracting workers.

This perspective generates a clear empirical prediction. If greenspace enhances quality of life, then equilibrium wages should be lower in counties with more greenspace, holding productivity constant and controlling for rents. The size of this compensating wage differential reflects the implicit value workers place on greenspace in their location decisions. Our empirical analysis tests this prediction by estimating how wages vary with urban greenspace across 950 U.S. urban counties.

4.2 Empirical Model

Using equation (2), we examine the relationship between urban greenspace and wages across counties in the United States from 2011 to 2019.

$$\log(\text{wage}_{ct}) = \beta_0 + \beta_1 \log(\text{NDVI}_{ct}) + \gamma_i \text{Controls}_{ct} + \theta_c + \tau_t + \varepsilon_{ct} \quad (2)$$

The dependent variable of the model is the logarithm of average annual wages for county c in year t , measured in USD. We also estimate alternative specifications, varying the dependent variable using the average annual employment level in county c in year t , and total population. Greenspace is represented by the NDVI value for county c in year t , ranging from 0 to 1, and serves as the primary variable of interest, making β_1 the coefficient of concern. Controls_{ct} is a vector of control variables, including geographic characteristics, socio-economic indicators, and industry composition, as discussed above. θ_c represents state-level fixed effects, while τ_t captures year fixed effects.

4.3 Identification

One of the primary concerns with identifying the impact of greenspace on wages is that places with higher wages may invest in more greenspace resulting in possible reverse causality. Greenspace may also be correlated with unobserved factors resulting in endogeneity. Furthermore, there can be simultaneous effects on wages and greenspace due to factors like government zoning policies, which might make our estimators biased. To address these concerns, we employ an IV approach utilizing historical land cover data and controlling for historical industrialization. Specifically, we leverage land cover data from the 1970s-1980s, sourced from the

United States Geological Survey (USGS), which predates our study period by approximately three decades. This temporal distance is crucial for our identification strategy.

$$\log(\widehat{\text{NDVI}}_{ct}) = \alpha_0 + \alpha_1 \text{HistLC}_{ct} + \alpha_2 \text{HistIC}_{ct} + \gamma_i \text{Controls}_{ct} + \theta_c + \tau_t + \varepsilon_{ct} \quad (3)$$

Equation (6) specifies the first stage of an IV approach, where the dependent variable is the predicted log of NDVI, denoted as $\log(\widehat{\text{NDVI}}_{ct})$, for county c in year t . This formulation leverages historical variables to isolate exogenous variation in urban greenspace, inspired from previous work (Stephens and Partridge, 2011). The term HistLC_{ct} represents historical land cover characteristics, such as forest or shrub density, which shape current greenspace levels but are plausibly uncorrelated with contemporary labor market shocks. HistIC_{ct} captures the historical industry composition of the county, reflecting long-run economic patterns that may influence the development and preservation of greenspace. Controls_{ct} includes additional observable county-level controls such as demographic, climatic, or spatial characteristics similar to the primary specification. The fixed effects θ_c and τ_t account for unobserved county-specific heterogeneity and common temporal shocks, respectively, and ε_{ct} is the error term. This first-stage regression generates exogenous variation in NDVI used in the second-stage wage equation to identify the causal effect of greenspace on labor market outcomes.

The selection of this instrument is predicated on two key assumptions. First, regarding relevance, we see a strong correlation between historic land cover patterns and contemporary NDVI values as evidenced by the first stage F-statistics (*see appendix 2*). Concerning the exclusion restriction, land cover from three decades prior is unlikely to exert a direct influence on current wage levels, except through its impact on present-day greenspace as measured by NDVI, when controlling for historical industrial composition. To test this, we regress wages on historic land cover and find no statistical significance. These results are shown in *Appendix 1*.

5 Results

5.1 Primary Results

The results in *table 2* provide empirical support for the hypothesized relationship between urban greenspace and labor market outcomes. Detailed regression results are provided in *Appendix 3*. *Column (1)* reports a simple OLS estimate, suggesting a statistically insignificant association between NDVI and wages. This likely reflects the influence of omitted variables or measurement error in the greenspace measure. Using our

preferred IV approach in *column (2)*, we find a statistically significant and negative relationship between greenspace and wages. Specifically, a 1% increase in NDVI is associated with a 0.064% decrease in wages, holding other factors constant. This aligns with predictions from the Roback model, where workers accept lower wages in exchange for higher amenity value from greenspace.

Table 2: Regression Results

VARIABLES	OLS	IV-Annual Avg Wage	IV-Squared Wage	IV-Avg Emp Level	IV-Population
Log Greenspace (NDVI)	0.000781 (0.000689)	-0.0677*** (0.0161)	-0.200*** (0.0456)	20,953*** (5045)	7.97e-08*** (6.51e-09)
Log Squared Greenspace (NDVI)			-0.052*** (0.0112)		
Observations	8,970	8,970	8,970	8,970	8,970
R-squared	0.900	0.859	0.859	0.986	0.986
Number of Counties	950	950	950	950	950
State FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
<i>First Stage:</i>		NDVI	NDVI	NDVI	NDVI
Historic Land Cover		0.090*** (0.0065)	0.090*** (0.0065)	0.090*** (0.0065)	0.090*** (0.0065)
CD Wald F-Stat		183.02	183.02	183.02	183.02

Standard errors in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Column (3) extends the analysis by including the squared term of NDVI to test for potential nonlinearities. Both the linear and squared terms are statistically significant and negative, indicating that an increasing value of greenspace increases the higher the wage compensation the workers are willing to get.

Beyond wages, we also examine broader economic effects. *Column (4)* shows that greater greenspace is associated with higher average employment levels, with a 1% increase in NDVI linked to approximately 17,500 more employed individuals. This is consistent with economic theory, where a lower wage would lead to cheaper cost of labor hiring and as a result, more workers will be employed. *Similarly, column (5)* reports a strong positive association with total population, suggesting that greener counties attract more residents. These findings imply that greenspace not only influences individual preferences as captured by wages but may also contribute to local economic vitality by increasing labor supply and population.

The first-stage F-statistics well exceed the conventional threshold, supporting the strength of the instrument (historic land cover) used to isolate exogenous variation in greenspace. Present NDVI is highly correlated with past land cover patterns of greenery, when controlling for present and historical industrial composition. The detailed first stage regressions are provided in the *Appendix 2*.

5.2 Heterogeneity in Effect

Industry Concentration

We examine whether the wage–amenity trade-off from greenspace varies with local industrial structure. In a spatial-equilibrium setting, if greenspace is a consumption amenity, workers accept lower pay in greener places; the size of that concession may depend on how concentrated (specialized) a county’s economy is. For each county–year we compute a Herfindahl–Hirschman Index (HHI) from 2-digit NAICS employment shares (*equation (4)*), and mean-center it to aid interpretation and reduce collinearity (*equation (5)*).

$$\text{HHI}_{ct} = \sum_s \text{share}_{s,ct}^2, \quad (4)$$

$$\text{cHHI}_{ct} = \text{HHI}_{ct} - \overline{\text{HHI}}., \quad (5)$$

Regression model with HHI:

$$\ln w_{ct} = \beta_1 \ln \text{NDVI}_{ct} + \beta_2 \text{cHHI}_{ct} + \beta_3 (\ln \text{NDVI}_{ct} \times \text{cHHI}_{ct}) + X'_{ct} \delta + \lambda_{s(c)} + \tau_t + \varepsilon_{ct}, \quad (6)$$

where X_{ct} includes climate, demographics, and urbanity controls, $\lambda_{s(c)}$ are state fixed effects, and τ_t are year fixed effects. Because greenness may be endogenous, we estimate (6) by 2SLS, instrumenting $\ln \text{NDVI}_{ct}$ with $\ln \text{LCMean}_{ct}$ (historic land cover), with robust standard errors.

In the heterogeneity specification *equation (6)*, three coefficients summarize the relationship, as shown in *Table 3*, with detailed results given in *appendix 4*. First, at average concentration ($\text{cHHI} = 0$), the elasticity of wages with respect to greenness is negative and statistically significant ($\hat{\beta}_1 = -0.162$), consistent with a compensating-differentials interpretation. Second, the interaction between greenness and concentration is negative and significant ($\hat{\beta}_3 = -2.82$), indicating that the wage decrease becomes more pronounced as industrial concentration rises. Third, the main effect of concentration is small and negative ($\hat{\beta}_2 = -2.76$), implying slightly lower wages in more concentrated local economies, holding greenness and controls constant. Together, these results show that greener places have lower wages on average and that this compensating differential is stronger in more specialized local economies.

Table 3: HHI Regression Results

VARIABLES	IV
Log Greenspace (NDVI)	-0.167*** (0.0248)
Log Greenspace $\times$ HHI	-0.00045*** (0.00015)
HHI	-0.000488** (0.00019)
Observations	8,970
R-squared	0.657
Number of Counties	950
State FE	Yes
Year FE	Yes

Standard errors in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

6 Conclusion

This study provides compelling evidence that urban greenspace plays a critical role in shaping labor market outcomes through the lens of compensating differentials. By applying a spatial equilibrium framework and using NDVI as a proxy for greenspace across 950 urban U.S. counties from 2011 to 2019, the analysis shows that increased greenspace is associated with lower wages. This negative relationship, confirmed through an IV approach using historical land cover data, suggests that workers accept lower pay in exchange for the non-monetary benefits provided by greener environments. This tradeoff underscores the utility-enhancing role of greenspace in urban areas, consistent with the theoretical predictions of Roback (1982).

Importantly, the wage effects of greenspace are not uniform across regions. Our sectoral heterogeneity analysis reveals that the magnitude of compensating wage differentials depends on local economic structures. Counties with more concentrated employment bases experience more pronounced wage reductions in greener environments. At the same time, greater greenspace is positively associated with employment levels and population growth, indicating that while workers may earn less, greener cities attract more people and support greater labor demand. This signals that greenspace contributes to local economic vitality.

The results have significant policy relevance: unlike fixed amenities such as coastline or climate, greenspace is malleable and can be shaped through urban planning. Investments in tree planting, park development, and green infrastructure can enhance city livability, attract talent, and foster economic growth, even in the absence of direct productivity gains.

As urban populations continue to grow and climate-related stressors intensify, prioritizing greenspace should be viewed not merely as an environmental or aesthetic goal, but as a viable economic development strategy. By recognizing the compensatory value of greenery, policymakers can justify and design greenspace interventions that support inclusive, resilient urban economies. Future research could further unpack how the quality, accessibility, and spatial distribution of greenspace mediate these labor market effects, particularly across different socio-economic groups.

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Appendix

Appendix 1: IV testing regression

Table 1: NDVI regressed with Historical Landcover

VARIABLES	ln_NDVI	ln_wage
hist_landcover	0.0898*** (0.00876)	0.000768 (0.000723)
miles to nearest MSA	0.000550 (0.000729)	-0.00147*** (0.000260)
present_landcover	7.12e-08*** (9.94e-09)	-8.27e-09** (3.74e-09)
HPI	0.000242*** (5.50e-05)	7.87e-05*** (6.24e-06)
% working in-county	-0.00131** (0.000592)	0.00121*** (0.000141)
% High School Degree	-0.00326 (0.00238)	0.00180*** (0.000316)
% Bachelors Degree	-0.00514 (0.00342)	0.00423*** (0.000446)
% Graduate Degree	0.00154 (0.00406)	0.00265*** (0.000555)
% Employed	0.00133 (0.00193)	5.81e-05 (0.000264)
Median HH income (2010 USD)	1.32e-06 (1.13e-06)	1.56e-06*** (1.68e-07)
Total Population	-7.74e-08* (4.56e-08)	-5.02e-08*** (1.27e-08)
% Black	0.00155* (0.000840)	0.00233*** (0.000281)
% American Indian/Alaskan	0.00773* (0.00436)	0.00167 (0.00105)
% Asian	-0.00591* (0.00348)	0.00550*** (0.000861)
% Hawaii/Pacific	0.0850** (0.0433)	0.00583 (0.00402)
% Other race	-0.00381 (0.00269)	-5.35e-05 (0.000271)
% Multiracial	0.0215*** (0.00662)	0.000558 (0.000692)
Annual avg precipitation	0.00417 (0.200)	0.0206 (0.0138)
NAICS - Mine, Oil, Gas	0.000813 (0.00384)	0.0162*** (0.000487)
NAICS - Utilities	0.00150 (0.00717)	0.0121*** (0.00129)
NAICS - Construction	0.00799** (0.00323)	0.00907*** (0.000406)
NAICS - Manufacturing 1	-0.000303 (0.00206)	0.00300*** (0.000368)
NAICS - Manufacturing 2	0.00270 (0.00236)	0.00626*** (0.000399)
NAICS - Manufacturing 3	0.000394 (0.00151)	0.00612*** (0.000300)
NAICS - Wholesale	-0.00198	0.00390***

Continued on next page

Table 4 – continued from previous page

VARIABLES	ln_NDVI	ln_wage
	(0.00464)	(0.000630)
NAICS - Retail 1	0.00413	-0.00306***
	(0.00563)	(0.000728)
NAICS - Retail 2	0.00463	-0.00293***
	(0.00378)	(0.000548)
NAICS - Transport 1	-0.00169	0.00431***
	(0.00389)	(0.000723)
NAICS - Transport 2	-0.00522	-0.00106**
	(0.00436)	(0.000506)
NAICS - Information	-0.00239	0.0148***
	(0.00884)	(0.00117)
NAICS - Finance, Insurance	0.00244	0.0111***
	(0.00329)	(0.000607)
NAICS - Real Estate	-0.00571	-0.000235
	(0.0132)	(0.00153)
NAICS - Prof Science Tech	-0.00169	0.0108***
	(0.00387)	(0.000610)
NAICS - Management	0.00940*	0.00824***
	(0.00512)	(0.000631)
NAICS - Admin, Waste	-0.00696**	0.000119
	(0.00352)	(0.000415)
NAICS - Education	0.00371	0.00111*
	(0.00307)	(0.000604)
NAICS - Healthcare	0.00385**	6.77e-05
	(0.00184)	(0.000340)
NAICS - Arts, Entertainment	0.00613*	-0.00122*
	(0.00366)	(0.000736)
NAICS - Accommodation, Food	-0.00560*	-0.00673***
	(0.00295)	(0.000520)
NAICS - Other Services	-0.00933*	-0.000795
	(0.00489)	(0.000544)
20 hist_ind.value_70	5.26e-06***	1.49e-06**
	(1.88e-06)	(7.13e-07)
40 hist_ind.value_70	1.06e-05***	1.99e-06
	(3.73e-06)	(1.43e-06)
50 hist_ind.value_70	2.14e-05	-2.22e-05***
	(1.81e-05)	(7.17e-06)
70 hist_ind.value_70	7.15e-06	7.12e-07
	(1.03e-05)	(4.02e-06)
90 hist_ind.value_70	-6.21e-06***	-1.56e-06*
	(2.38e-06)	(9.13e-07)
100 hist_ind.value_70	-4.86e-05**	1.15e-05
	(2.43e-05)	(9.71e-06)
200 hist_ind.value_70	8.80e-06	-2.47e-06
	(6.18e-06)	(2.41e-06)
300 hist_ind.value_70	1.91e-07	1.52e-05***
	(4.85e-06)	(1.82e-06)
400 hist_ind.value_70	3.07e-06*	5.99e-07
	(1.76e-06)	(6.88e-07)
500 hist_ind.value_70	-1.56e-05***	3.42e-06**
	(4.54e-06)	(1.73e-06)
610 hist_ind.value_70	1.49e-05***	-3.86e-06**
	(4.09e-06)	(1.60e-06)
620 hist_ind.value_70	-1.78e-06	-4.59e-06***
	(3.94e-06)	(1.56e-06)
700 hist_ind.value_70	-5.12e-06	-4.19e-07
	(3.75e-06)	(1.48e-06)
910 hist_ind.value_70	-6.57e-06**	2.86e-06**

Continued on next page

Table 4 – continued from previous page

VARIABLES	ln_NDVI	ln_wage
	(3.02e-06)	(1.18e-06)
920 hist_ind_value_70	-4.51e-06**	-1.62e-06**
	(2.10e-06)	(8.12e-07)
Constant	-1.631***	9.978***
	(0.207)	(0.0387)
present_NDVI		-0.0146***
		(0.00331)
Observations	8,419	
Number of geoid	936	

Standard errors in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Appendix 2: First-Stage Regressions

VARIABLES	(1) First-stage: $\ln_yndvi$
$\ln_landcover_mean$	0.0794*** (0.00614)
$high_temp_days$	0.00311*** (0.000379)
$near_dist$	0.000441 (0.000566)
$present_landcover$	6.70e-08*** (9.06e-09)
$rent$	-0.0000116 (0.0000349)
$work_incounty$	-0.00147*** (0.000429)
$aland$	-3.86e-11*** (2.72e-12)
edu_perc_hs	-0.00202 (0.00184)
edu_perc_bach	-0.00537** (0.00235)
edu_perc_grad	0.00709** (0.00318)
$perc_employed$	0.00230 (0.00149)
pop_tot_pop	-1.03e-07*** (3.73e-08)
$race_perc_black$	0.000916* (0.000545)
$race_perc_amindnat$	0.0112*** (0.00351)

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VARIABLES	(1) First-stage: ln_yndvi
race_perc_asian	-0.00565* (0.00322)
race_perc_hawaii	0.0612 (0.0445)
race_perc_others	-0.00498** (0.00243)
race_perc_multirace	0.0189*** (0.00541)
precipitation	1.040*** (0.243)
naics2_21	-0.000613 (0.00321)
naics2_22	0.00173 (0.00530)
naics2_23	0.00731*** (0.00241)
naics2_31	0.00101 (0.00150)
naics2_32	0.00273 (0.00168)
naics2_33	0.000998 (0.00130)
naics2_42	0.000160 (0.00338)
naics2_44	0.0107*** (0.00408)
naics2_45	0.00213 (0.00241)
naics2_48	0.000719
<i>continued on next page</i>	

VARIABLES	(1) First-stage: $\ln_yndvi$
	(0.00259)
naics2_49	-0.00382
	(0.00269)
naics2_51	-0.00250
	(0.00650)
naics2_52	0.00376*
	(0.00220)
naics2_53	-0.00547
	(0.00891)
naics2_54	0.00199
	(0.00264)
naics2_55	0.00887***
	(0.00331)
naics2_56	-0.00322
	(0.00249)
naics2_61	0.00345
	(0.00244)
naics2_62	0.00240*
	(0.00138)
naics2_71	0.00606**
	(0.00243)
naics2_72	-0.00222
	(0.00151)
naics2_81	-0.01396**
	(0.00675)
hist_ind_value_7020	4.80e-06***
	(1.45e-06)
hist_ind_value_7040	0.0000126***
	(3.23e-06)

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VARIABLES	(1) First-stage: ln_yndvi
hist_ind_value_7050	0.0000153 (0.0000146)
hist_ind_value_7060	0 (omitted)
hist_ind_value_7070	0.0000111 (8.27e-06)
hist_ind_value_7080	0 (omitted)
hist_ind_value_7090	-6.69e-06*** (2.01e-06)
hist_ind_value_70100	-0.0000483** (0.0000195)
hist_ind_value_70200	8.71e-06* (5.00e-06)
hist_ind_value_70300	2.64e-06 (4.18e-06)
hist_ind_value_70400	3.63e-06** (1.49e-06)
hist_ind_value_70500	-0.0000119*** (4.26e-06)
hist_ind_value_70610	0.0000121*** (3.50e-06)
hist_ind_value_70620	-5.96e-07 (2.95e-06)
hist_ind_value_70700	-1.94e-06 (3.86e-06)
hist_ind_value_70800	0 (omitted)
hist_ind_value_70900	0 (omitted)
hist_ind_value_70910	-5.13e-06** (2.27e-06)
hist_ind_value_70920	-4.23e-06***

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VARIABLES	(1) First-stage: ln_yndvi
	(1.53e-06)
hist_ind_value_70930	0 (omitted)
<i>Year dummies</i>	
2011	0.2688*** (0.0220)
2012	-0.0909*** (0.0210)
2013	-0.1647*** (0.0235)
2014	0.3994*** (0.0187)
2015	0.2680*** (0.0199)
2016	0.3856*** (0.0186)
2017	0.3305*** (0.0232)
2018	0.5377*** (0.0216)
2019	0.4523*** (0.0202)
<i>State dummies (selected)</i>	
state_code 4	-0.2047* (0.1151)
state_code 5	0.1124** (0.0451)
state_code 6	0.2946*** (0.0598)
state_code 8	-0.1943***
<i>continued on next page</i>	

VARIABLES	(1) First-stage: ln_yndvi
	(0.0604)
state_code 10	-0.0462
	(0.0787)
state_code 12	-0.3664***
	(0.0432)
state_code 13	-0.0487*
	(0.0292)
state_code 16	0.0953
	(0.0747)
state_code 17	0.1091***
	(0.0385)
state_code 18	0.0890***
	(0.0341)
state_code 19	0.1297***
	(0.0443)
state_code 20	0.3889***
	(0.0394)
state_code 21	0.2226***
	(0.0318)
state_code 23	-0.0983
	(0.0677)
state_code 24	0.0142
	(0.0449)
state_code 25	-0.0578
	(0.0528)
state_code 26	0.2070***
	(0.0356)
state_code 27	0.1459***
	(0.0486)
<i>continued on next page</i>	

VARIABLES	(1) First-stage: ln_yndvi
state_code 28	0.0521 (0.0400)
state_code 29	0.2767*** (0.0350)
state_code 30	0.0632 (0.0883)
state_code 31	0.2568*** (0.0543)
state_code 32	-0.5109*** (0.1565)
state_code 33	-0.0281 (0.0835)
state_code 34	-0.0467 (0.0462)
state_code 35	-0.7966*** (0.0919)
state_code 36	-0.0172 (0.0449)
state_code 37	0.1019*** (0.0278)
state_code 38	0.2048*** (0.0784)
state_code 39	0.0299 (0.0360)
state_code 40	0.1622*** (0.0550)
state_code 41	0.0656 (0.0613)
state_code 42	-0.0752**
<i>continued on next page</i>	

VARIABLES	(1) First-stage: ln_yndvi
	(0.0382)
state_code 44	0.0166
	(0.0533)
state_code 45	0.0938***
	(0.0305)
state_code 46	0.2268***
	(0.0565)
state_code 47	0.1023***
	(0.0288)
state_code 48	0.1862***
	(0.0328)
state_code 49	0.1173*
	(0.0674)
state_code 50	-0.0196
	(0.0740)
state_code 51	0.1073***
	(0.0330)
state_code 53	-0.1312*
	(0.0680)
state_code 54	0.0777**
	(0.0376)
state_code 55	0.2499***
	(0.0398)
state_code 56	0.3168***
	(0.0888)
Constant	-1.901***
	(0.165)
Observations	8,970
State FE	Yes
<i>continued on next page</i>	

VARIABLES	(1) First-stage: ln_yndvi
Year FE	Yes
<i>F test of excluded instruments: $F(1, 8859) = 167.15$, $Prob > F = 0.0000$</i>	
<i>Sanderson–Windmeijer multivariate F: $F(1, 8859) = 167.15$, $Prob > F = 0.0000$</i>	
<i>Notes:</i> Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.	

Appendix 3: Second-Stage Primary Regressions

VARIABLES	(1) Wage	(2) Pop	(3) Emp	(4) Square
ln_yndvi	-0.0677*** (0.0161)	7.97e-08*** (6.51e-09)	20,953*** (5,045)	-0.200*** (0.0456)
ln_yndvi_sq				-0.0527*** (0.0120)
miles to nearest MSA	-0.000226** (0.000113)	-2.97e-10*** (0)	14.16 (35.75)	-0.000212* (0.000116)
present_landcover	-4.27e-09** (1.82e-09)	-0*** (0)	0.00273*** (0.000705)	-8.42e-09*** (1.21e-09)
% of people working in-county	0.00122*** (8.88e-05)	-2.75e-10*** (0)	510.6*** (31.81)	0.00134*** (9.19e-05)
County land area (sq km)	-0 (0)	0*** (0)	-1.66e-06*** (3.13e-07)	0** (0)
% of population with Highschool Degree	0.00120*** (0.000395)	-8.86e-10*** (1.54e-10)	-1,271*** (141.6)	0.000772* (0.000431)
% of population with Bachelors Degree	0.00488*** (0.000567)	-1.23e-10 (1.97e-10)	-1,554*** (207.3)	0.00497*** (0.000576)
% of population with Graduate Degree	0.00605*** (0.000765)	-1.84e-09*** (2.76e-10)	-1,403*** (296.4)	0.00517*** (0.000791)
% of people employed	0.00108*** (0.000318)	0 (1.22e-10)	708.4*** (100.8)	0.000849** (0.000331)
Total Population	-3.35e-08*** (5.91e-09)	1*** (0)	0.388*** (0.00681)	-2.40e-08*** (6.17e-09)
% of people Black	0.00257*** (0.000121)	-1.04e-10** (0)	-295.2*** (50.08)	0.00237*** (0.000126)

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VARIABLES				(1) Wage	(2) Pop	(3) Emp	(4) Square
% of people American In-				0.00201***	-7.01e-10**	262.9	0.000785
dian/Alaskan				(0.000594)	(2.91e-10)	(184.0)	(0.000600)
% of people Asian				0.00417***	3.22e-10	3,112***	0.00453***
				(0.000668)	(2.70e-10)	(386.9)	(0.000643)
% of people Hawaii/Pacific				0.0369***	-4.11e-09	-7,268**	0.0391***
				(0.00965)	(3.57e-09)	(3,263)	(0.0101)
% of people other race				0.00204***	-3.14e-10	-1,618***	0.00232***
				(0.000443)	(2.02e-10)	(208.1)	(0.000443)
% of people multiracial				0.00903***	-1.77e-09***	-1,445***	0.00836***
				(0.00152)	(4.59e-10)	(392.8)	(0.00147)
Annual average precipitation				0.189***	-1.47e-07***	-9,480	0.169***
				(0.0460)	(2.12e-08)	(15,927)	(0.0454)
NAICS - Mine,Oil,Gas				0.0165***	-6.24e-10**	-140.0	0.0166***
				(0.000592)	(2.49e-10)	(184.9)	(0.000601)
NAICS - Utilities				0.0256***	-2.59e-10	217.5	0.0255***
				(0.00108)	(4.37e-10)	(244.5)	(0.00110)
NAICS - Construction				0.00550***	-2.08e-09***	-767.7***	0.00539***
				(0.000604)	(2.11e-10)	(160.6)	(0.000616)
NAICS - Manufacturing 1				0.00182***	-4.60e-10***	21.65	0.00172***
				(0.000362)	(1.22e-10)	(84.23)	(0.000363)
NAICS - Manufacturing 2				0.0106***	-3.70e-10***	-195.9**	0.0105***
				(0.000418)	(1.41e-10)	(80.46)	(0.000421)
NAICS - Manufacturing 3				0.00911***	-1.66e-10	146.7**	0.00913***
				(0.000338)	(1.02e-10)	(65.40)	(0.000356)
NAICS - Wholesale				0.00504***	2.56e-10	597.7***	0.00533***
				(0.000774)	(2.76e-10)	(209.6)	(0.000778)
NAICS - Retail 1				-0.00652***	-2.43e-09***	-3,247***	-0.00695***
				(0.00101)	(3.49e-10)	(255.6)	(0.00102)

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VARIABLES	(1) Wage	(2) Pop	(3) Emp	(4) Square
NAICS - Retail 2	-0.00263*** (0.000750)	-9.96e-10*** (2.07e-10)	-1,122*** (168.8)	-0.00235*** (0.000758)
NAICS - Transport 1	0.00814*** (0.000503)	-7.98e-10*** (2.25e-10)	759.6*** (156.9)	0.00788*** (0.000524)
NAICS - Transport 2	0.00349*** (0.000587)	3.88e-10 (2.45e-10)	194.3 (157.5)	0.00345*** (0.000599)
NAICS - Information	0.0133*** (0.00182)	9.87e-10* (5.43e-10)	7,196*** (840.3)	0.0128*** (0.00181)
NAICS - Finance, Insurance	0.00995*** (0.000716)	-3.12e-10* (1.77e-10)	294.4* (162.0)	0.00970*** (0.000745)
NAICS - Real Estate	0.00371 (0.00234)	4.92e-09*** (8.04e-10)	644.0 (548.2)	0.00276 (0.00238)
NAICS - Prof Science Tech	0.0211*** (0.000622)	-3.32e-10 (2.14e-10)	3,096*** (331.9)	0.0206*** (0.000641)
NAICS - Management	0.0253*** (0.00126)	-1.17e-09*** (2.86e-10)	1,666*** (310.9)	0.0252*** (0.00127)
NAICS - Admin, Waste	0.00156** (0.000645)	2.06e-10 (2.06e-10)	-33.11 (166.4)	0.00194*** (0.000639)
NAICS - Education	0.000496 (0.000382)	-6.23e-10*** (2.00e-10)	-255.7* (142.6)	0.000237 (0.000380)
NAICS - Healthcare	0.00268*** (0.000366)	-5.73e-11 (1.16e-10)	-932.4*** (95.53)	0.00250*** (0.000368)
NAICS - Arts, Entertainment	-0.000912 (0.000689)	-1.08e-09*** (2.19e-10)	266.8 (182.0)	-0.000939 (0.000660)
NAICS - Accommodation,Food	-0.00362*** (0.000411)	3.49e-10*** (1.26e-10)	-105.1 (81.38)	-0.00373*** (0.000410)
NAICS - Other Services	-0.0119*** (0.000961)	9.60e-10* (5.37e-10)	-85.72 (250.1)	-0.0105*** (0.00126)
20 value_70	6.25e-07**	-0	-0.0602	2.21e-07

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VARIABLES	(1) Wage	(2) Pop	(3) Emp	(4) Square
	(2.63e-07)	(0)	(0.193)	(2.58e-07)
40 value_70	4.17e-07	-0***	-1.358***	-4.27e-07
	(6.23e-07)	(0)	(0.399)	(6.01e-07)
50 value_70	-7.23e-06***	0***	-6.974***	-9.16e-06***
	(2.23e-06)	(0)	(1.066)	(2.24e-06)
70 value_70	-1.57e-06	-0***	3.358***	-1.91e-06
	(1.30e-06)	(0)	(0.776)	(1.27e-06)
90 value_70	-1.48e-07	0***	-1.038***	5.96e-07*
	(3.35e-07)	(0)	(0.296)	(3.46e-07)
100 value_70	1.19e-05***	0***	-7.509***	1.07e-05***
	(3.08e-06)	(0)	(1.796)	(3.10e-06)
200 value_70	2.80e-06***	-0	1.450**	-4.51e-07
	(7.41e-07)	(0)	(0.671)	(9.25e-07)
300 value_70	5.70e-06***	-0***	10.41***	6.23e-06***
	(6.56e-07)	(0)	(0.749)	(7.22e-07)
400 value_70	3.43e-07	-0***	0.289	-2.21e-07
	(2.32e-07)	(0)	(0.280)	(2.58e-07)
500 value_70	2.03e-07	0	-2.847***	1.78e-06**
	(7.91e-07)	(0)	(0.535)	(7.81e-07)
610 value_70	-8.55e-07	-0***	5.845***	-2.79e-06***
	(5.62e-07)	(0)	(0.567)	(6.10e-07)
620 value_70	-4.36e-06***	-0***	0.609	-4.70e-06***
	(5.52e-07)	(0)	(0.563)	(5.77e-07)
700 value_70	-7.89e-07	-0**	5.898***	-1.03e-06
	(5.95e-07)	(0)	(0.714)	(6.61e-07)
910 value_70	2.16e-06***	0	-1.508***	2.54e-06***
	(3.95e-07)	(0)	(0.265)	(3.86e-07)
920 value_70	-7.62e-07***	0	0.390*	-5.82e-07**
	(2.75e-07)	(0)	(0.206)	(2.77e-07)

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VARIABLES	(1) Wage	(2) Pop	(3) Emp	(4) Square
Constant	9.675*** (0.0431)	2.67e-07*** (1.87e-08)	100,066*** (13,971)	9.650*** (0.0473)
Observations	8,970	8,970	8,970	8,970
R-squared	0.859	0.986	0.986	0.858
<i>Notes:</i> Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.				

Appendix 4: HHI Interaction Regression

VARIABLES	(1) HHI
NDVI	-0.16967*** (0.02483)
NDVI $\times$ HHI	-0.000450*** (0.000151)
HHI	-0.000488** (0.000190)
Annual average precipitation	0.353*** (0.0739)
miles to nearest MSA	0.000535*** (0.000169)
% of people working in-county	0.000260* (0.000153)
County land area (sq km)	-0*** (0)
% of population with Bachelors Degree	-0.00108 (0.00116)
% of population with Graduate Degree	0.0110*** (0.00119)
% of people Black	0.00251*** (0.000183)
% of people American Indian/Alaskan	0.00480*** (0.00105)
% of people Asian	0.00228*** (0.000781)
% of people Hawaii/Pacific	0.0210*

continued on next page

VARIABLES	(1) HHI
	(0.0114)
% of people other race	-0.00169*
	(0.000965)
% of people multiracial	0.00930***
	(0.00200)
NAICS - Mine,Oil,Gas	0.0192***
	(0.000898)
NAICS - Utilities	0.0263***
	(0.00232)
NAICS - Construction	0.00912***
	(0.000846)
NAICS - Manufacturing 1	0.00313***
	(0.000679)
NAICS - Manufacturing 2	0.0113***
	(0.000812)
NAICS - Manufacturing 3	0.00725***
	(0.000958)
NAICS - Wholesale	0.00818***
	(0.00114)
NAICS - Retail 1	-0.00532***
	(0.00145)
NAICS - Retail 2	-0.00145
	(0.000894)
NAICS - Transport 1	0.00649***
	(0.00123)
NAICS - Transport 2	0.000318
	(0.00160)
NAICS - Information	0.0144***
	(0.00244)
<i>continued on next page</i>	

VARIABLES	(1) HHI
NAICS - Finance, Insurance	0.0112*** (0.000769)
NAICS - Real Estate	0.000211 (0.00348)
NAICS - Prof Science Tech	0.0206*** (0.00104)
NAICS - Management	0.0229*** (0.00152)
NAICS - Admin, Waste	-0.000579 (0.00104)
NAICS - Education	0.000746 (0.000883)
NAICS - Healthcare	0.00176*** (0.000565)
NAICS - Arts, Entertainment	0.00390*** (0.00118)
NAICS - Accommodation,Food	-0.00603*** (0.00109)
NAICS - Other Services	-0.0102*** (0.00202)
20 value_70	-2.64e-07 (4.63e-07)
40 value_70	9.99e-08 (9.15e-07)
50 value_70	-1.44e-06 (4.95e-06)
70 value_70	-6.44e-06** (2.85e-06)
90 value_70	1.19e-06**
<i>continued on next page</i>	

VARIABLES	(1) HHI
	(5.78e-07)
100 value_70	9.45e-06 (6.23e-06)
200 value_70	-1.08e-07 (1.41e-06)
300 value_70	1.79e-06 (1.14e-06)
400 value_70	-4.33e-07 (4.11e-07)
500 value_70	-1.39e-07 (1.04e-06)
610 value_70	-1.79e-07 (1.02e-06)
620 value_70	-5.66e-06*** (8.76e-07)
700 value_70	4.77e-07 (1.02e-06)
910 value_70	1.11e-06 (7.47e-07)
920 value_70	1.40e-07 (5.66e-07)
Constant	5.205*** (0.0530)
Observations	8,970
R-squared	0.657

Notes: Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.